



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-14

Experiment 14. Decision Tree

Aim

To implement a Decision Tree classification algorithm using Python.

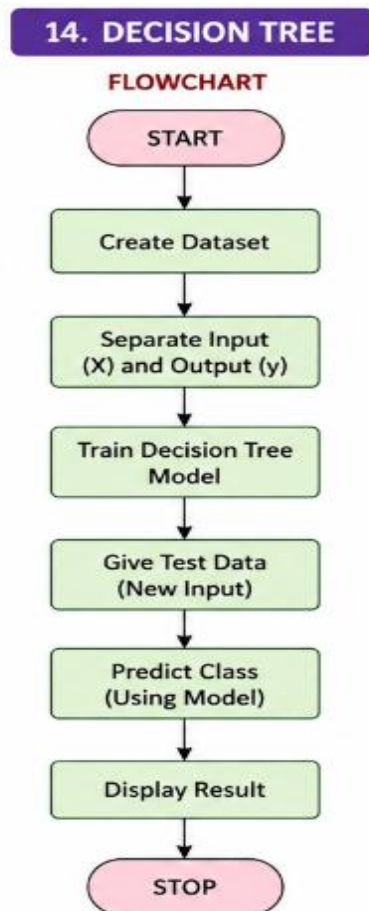
Objectives

- To understand Decision Tree classification.
- To train a model using sample data.
- To predict the class of new data.

Algorithm

1. Import the Decision Tree classifier.
2. Create the input and output data.
3. Train the Decision Tree model.
4. Give test data.
5. Predict the class.
6. Display the result.

Flowchart



Code

```
from sklearn.tree import
```

```
DecisionTreeClassifier
```

```
X = [[20, 30000],
```

```
      [25, 40000],
```

```
      [30, 50000],
```

```
[35, 60000],
```

```
[40, 70000]]
```

```
y = ["No", "No", "Yes",
```

```
"Yes", "Yes"]
```

```
model =
```

```
DecisionTreeClassifier()
```

```
model.fit(X, y)
```

```
age = int(input("Enter age:
```

```
"))
```

```
salary = int(input("Enter
```

```
salary: "))result =
```

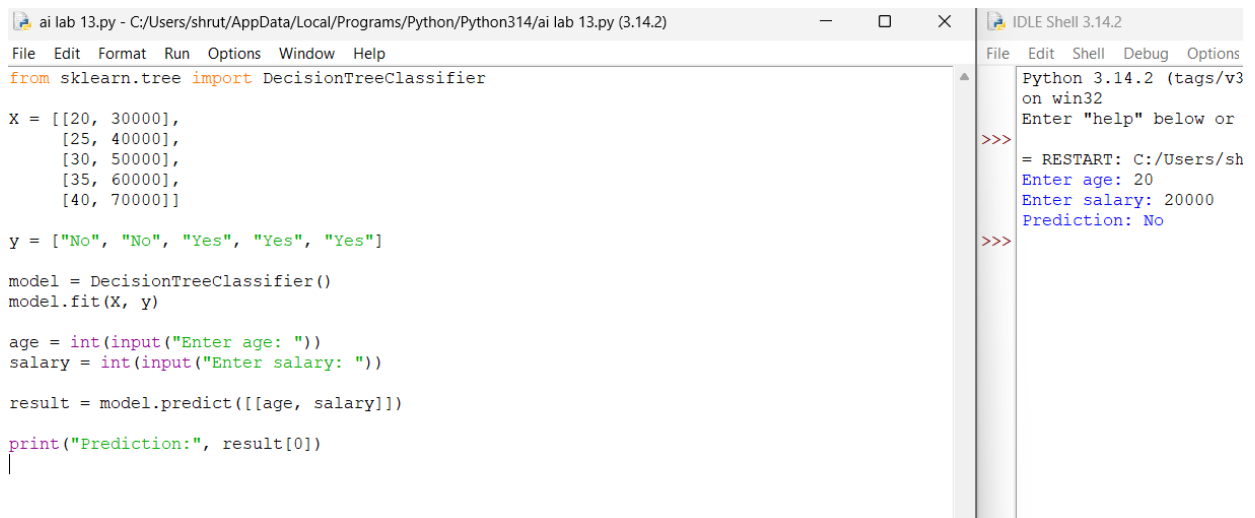
```
model.predict([[age,
```

```
salary]])
```

```
print("Prediction:",
```

```
result[0])
```

Output



```
ai lab 13.py - C:/Users/shrut/AppData/Local/Programs/Python/Python314/ai lab 13.py (3.14.2)
File Edit Format Run Options Window Help
from sklearn.tree import DecisionTreeClassifier

X = [[20, 30000],
      [25, 40000],
      [30, 50000],
      [35, 60000],
      [40, 70000]]

y = ["No", "No", "Yes", "Yes", "Yes"]

model = DecisionTreeClassifier()
model.fit(X, y)

age = int(input("Enter age: "))
salary = int(input("Enter salary: "))

result = model.predict([[age, salary]])

print("Prediction:", result[0])
|

IDLE Shell 3.14.2
File Edit Shell Debug Options
Python 3.14.2 (tags/v3.14.2:2b50132c, Jun 11 2024) on win32
Enter "help" below or
>>>
= RESTART: C:/Users/shrut/AppData/Local/Programs/Python/Python314/IDLE Shell
Enter age: 20
Enter salary: 20000
Prediction: No
>>>
```

Result

Prediction: Yes

Conclusion

Thus, the Decision Tree classification algorithm was successfully implemented using Python.